

Directional Partial-Wave Sweeps and the Krylov Solve

The propagator matvec for elastic multiple scattering on a depth-stratified voxel grid

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Abstract

The multiple scattering of elastic waves by voxels superimposed on a depth-stratified background is a Foldy–Lax / volume integral equation, $T = T_0(I - G_0T_0)^{-1}$, solved by a Krylov method whose only nontrivial operation is the application of the background propagator G_0 . This note describes that application as an alternating-direction partial-wave summation: the lateral coupling is summed by constant- z left-right (k_x) and constant- (z, x) in-out (k_y) cumulative-phase sweeps, and the coupling between depth planes by the up-down Kennett reflectivity recursion (k_z). Choosing the lateral directions for the in-plane coupling sums the propagator along axes in which the voxels are separated, so the k_z partial-wave sum that diverges at zero depth separation is never evaluated. The sweeps are a pure forward matvec; all multiple scattering, to every order, is resolved by the Krylov iterations.

Companion series. *One of a companion series on elastic multiple scattering from cubic heterogeneities on a depth-stratified background, the T -matrix programme $T = T_0(I - G_0T_0)^{-1}$: (I) the stratified inter-site propagator, the same-depth convergence theory, the SH and P/SV traction kernels, interplane side scattering and the first-shell near stencil (`EwaldIntraPlanePropagator`); (II) the contrast-source volume integral equation and the CG-FFT / seismic grid T -matrix lineage (`ContrastSourceVIE`); (III) the directional partial-wave sweep matvec and the Krylov multiple-scattering solve (`DirectionalSweepSolver`); with the empirical companion `RQ1_Planar-Decomposition-Proposal`. This is note (III).*

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1 The system, and where the cost lives

Voxels (cubic cells) are superimposed on a background that carries the lateral average of the medium at each depth: laterally homogeneous within a constant- z plane, layered in z . Each voxel has an on-site T-matrix T_0 , the single-cell scatterer (Eshelby with the $r = 0$ delta-function correction up to the 27-component Galerkin form, the self-term already closed), and the voxels are coupled by the background propagator G_0 . The multiple scattering is the Foldy–Lax system [1, 2, 3]

$$(I - G_0 T_0) \psi = \psi^{\text{inc}}, \quad T = T_0 (I - G_0 T_0)^{-1}, \quad (1)$$

where ψ collects the exciting (incoming) fields at the voxels. The operator T_0 is block-diagonal and local. The whole cost is the off-diagonal G_0 : the inter-voxel communication. The strategy is to solve (1) by a Krylov method whose matrix-vector product is one application of G_0 , and to make that application fast.

The essential point is the division of labour. The application of G_0 is a *forward* operation, a summation of partial waves that carries the radiation of every voxel to every other. It contains no inversion and no embedding. All of the multiple scattering, the reverberations to every order, is built and resummed by the Krylov iterations, not by the propagator.

2 The matvec is a partial-wave summation

Given source amplitudes $b = T_0 \psi$ at the voxels, the matvec returns the field they radiate, $\psi' = G_0 b$. For the homogeneous part of the background the propagator is a sum of partial waves, and a partial wave’s phase accumulates linearly in space: e^{ikx} at the next voxel is the current one times e^{ikp} , with p the pitch. The sum is therefore built by carrying that running phase along a line, one term added per voxel. This is the sweep. The only freedom is which axis to sum along, and the choice is dictated by where the voxels are separated.

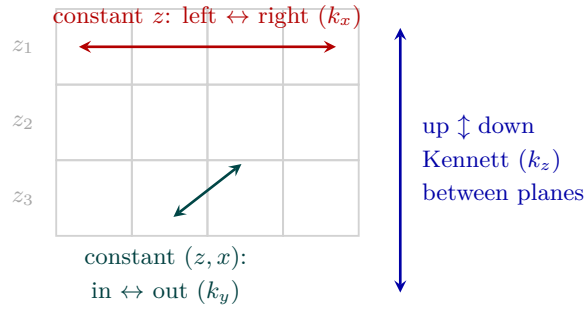


Figure 1: The G_0 matvec as three directional partial-wave sweeps over the voxel grid (cross-section; the third axis y runs into the page). Within each depth plane the lateral coupling is summed by constant- z left-right (k_x) and constant- (z, x) in-out (k_y) cumulative-phase sweeps; the coupling between depth planes is the up-down Kennett reflectivity recursion (k_z). The lateral directions carry the in-plane coupling along axes of nonzero separation, so the k_z sum is never evaluated at $\Delta z = 0$.

2.1 Lateral sweeps: constant- z left-right and constant- (z, x) in-out

Within a depth plane the background is homogeneous, so the lateral propagation is pure phase, with no background reflections to sum. Split the in-plane propagator on the k_x pole into a right-going and a left-going partial wave and sum each by a sweep in its own direction. With a^+ the right-going accumulated field, s_n the source radiated at voxel n , and $\Phi_x = e^{ik_x p}$ the

one-pitch lateral propagator,

$$a_{n+1}^+ = (a_n^+ + s_n) \Phi_x \quad (\text{left to right}), \quad a_{n-1}^- = (a_n^- + s_n) \Phi_x \quad (\text{right to left}). \quad (2)$$

Each partial wave is carried along the direction in which it decays: the right-going evanescent branch $e^{-\kappa_x x}$ is swept to the right, the left-going branch to the left, so the growing exponential is never formed and no reflection/transmission bookkeeping is needed for stability. The in-out sweep in y is identical, run along each constant- (z, x) line on the k_y pole. The two together sum the full in-plane partial-wave series as nested one-dimensional phase accumulations.

2.2 Vertical sweep: up-down Kennett between depth planes

In z the background is not homogeneous: the lateral average changes from plane to plane, so the depth-average layering has its own reflections and transmissions. The coupling between depth planes is therefore the up-down Kennett reflectivity recursion [4] on the k_z pole, propagating the source amplitudes through the layered background and summing its internal up/down multiples. This is the one sweep that carries reflection and transmission coefficients, and it does so for the *background* layering, not for the heterogeneity; it is the standard invariant-embedding recursion, stable by construction, and its coefficients depend only on the background and are formed once. The heterogeneity scattering remains entirely in T_0 and the Krylov loop.

2.3 Why these directions

The conventional construction sums the in-plane coupling on the k_z pole, the up/down decomposition, whose surviving (k_x, k_y) integral loses its convergence factor $e^{-\kappa_\perp |\Delta z|}$ at $\Delta z = 0$ and diverges at equal depth. The lateral split avoids it outright: the in-plane coupling is summed on the k_x and k_y poles, along which distinct in-plane voxels are separated, so the divergent same-depth k_z sum is never the one evaluated. The k_z sweep is used only between planes, where $\Delta z \neq 0$ and the Kennett recursion is exact and stable. Every sweep marches along an axis of nonzero separation.

2.4 The same-depth term in a stratified background

The lateral sweep of §2.1 sums the same-depth coupling of the *whole space*. That is not the whole of it. When the background is stratified, two voxels at the same depth are also coupled by paths that leave one voxel, reflect from a layer boundary above or below, and return to the plane. Those paths are absent from the lateral sweep by construction, because the k_x residue carries only the direct term.

They cannot simply be added by evaluating the stratified propagator at $\Delta z = 0$ and integrating: that is the divergence of §2.3, and it is present in the stratified kernel exactly as in the whole-space one, its strain-strain block growing like $|k_x|$. The resolution is to subtract it. Writing G_0^{lay} for the stratified propagator between a plane and itself and G_0^∞ for the whole-space kernel at equal depth,

$$G_0^{\text{lay}}(\Delta z = 0) = \underbrace{G_0^\infty(\Delta z = 0)}_{\text{lateral sweep, by } k_x \text{ residue}} + \underbrace{\Delta G_0}_{\text{reverberation only}}, \quad (3)$$

and the two terms on the right are individually non-integrable while their difference is not. Every path contributing to ΔG_0 travels at least twice the distance H to the nearest interface, so it carries $e^{ik_z 2H}$ and ΔG_0 decays exponentially in k_x . Its k_x integral therefore converges, and the divergence cancels between the two terms rather than being regularised in either.

Two consequences are worth stating plainly. First, ΔG_0 vanishes identically when the background is uniform, so the construction reduces to the whole-space sweep with nothing left

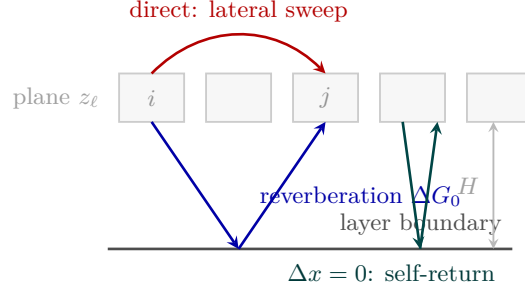


Figure 2: The same-depth interactions of (3), at one depth plane. The **direct** whole-space coupling between voxels i and j is summed by the lateral sweep, on the k_x pole, exactly. The **reverberation** leaves the plane, reflects from a layer boundary a distance H away and returns to it; the lateral sweep cannot carry it, and it is what ΔG_0 supplies. Both terms diverge on their own — each grows like $|k_x|$ — and their difference does not, because every reverberating path travels at least $2H$ and so carries $e^{ik_z 2H}$. The **third path** is the same process returning to the *source voxel*: it is a genuine self-interaction, it is not contained in T_0 (which is the single-site T-matrix of the whole space), and dropping it — the natural move, since the whole-space self-term is excluded everywhere else — would silently discard every layer-return-to-self.

over. Second, ΔG_0 is retained at $\Delta x = 0$: a voxel *does* interact with itself through the layering, and since T_0 is the single-site T -matrix of the whole space, its self-term does not contain that path. Dropping the $\Delta x = 0$ entry — the natural thing to do, since the whole-space self-term is excluded everywhere else — would silently discard every layer-return-to-self.

3 Alternating-direction composition

The three sweep pairs compose by alternating direction, in the spirit of the alternating-direction methods of Peaceman and Rachford [5]: the two-dimensional in-plane sum factors into an x pass and a y pass, and the full three-dimensional propagator into those plus the z recursion. Each pass is a one-dimensional, $O(N)$ accumulation, and the matvec is their ordered application:

1. **up-down (k_z , Kennett)**: propagate the source amplitudes between depth planes through the layered background;
2. **left-right (k_x)**: along each constant- z row, sweep left to right for the right-going field and right to left for the left-going field;
3. **in-out (k_y)**: along each constant- (z, x) line, sweep in and out the same way.

The result is $\psi' = G_0 b$, the field every voxel radiates onto every other, summed along directions of nonzero separation throughout. Six direction-pure sweeps in all, none of which ever forms the $\Delta z = 0$ sum.

4 The Krylov solve

The matvec of Section 3 is a pure forward application of G_0 . The multiple scattering is solved around it by a Krylov method (GMRES [6]) applied to (1). One iteration is

$$v \mapsto v - G_0 T_0 v, \quad (4)$$

that is, one block-diagonal T_0 (local, the cube T-matrix) followed by one G_0 (the three sweeps). The Krylov subspace builds the scattering series and resums it, reaching the solution in far fewer applications than the bare Neumann (Born) series and remaining stable at strong contrast

where that series would diverge. The reverberations and back-and-forth that the forward sweep deliberately does *not* contain are produced here, across the iterations.

For strong contrast the iteration is preconditioned, the role played by the renormalised scattering series of Jakobsen and Wu [7] and the convergent Born series of Osnabrugge, Leedum-rongwattanakun and Vellekoop [8]: a preconditioner on the Krylov solve, transparent to the sweep, which never needs to know the contrast level.

The stack, end to end.

1. **Setup.** Form T_0 per voxel (self-term closed) and the background Kennett reflectivity coefficients (once).
2. **Matvec** $G_0 b$. Up-down Kennett between planes, then left-right (k_x) and in-out (k_y) cumulative-phase sweeps within planes.
3. **Solve.** GMRES on $(I - G_0 T_0)\psi = \psi^{\text{inc}}$, preconditioned at strong contrast, each step one T_0 and one matvec.
4. **Output.** The exciting fields ψ , hence the global T and any observable (reflection matrix, shot gather) by one more radiation of $b = T_0 \psi$.

5 Validation status

The 2½-D case (heterogeneity in (z, x) , y invariant, k_y a parameter) is implemented and gated. Each claim below is the same physics through two independent code paths, so the residual measures defect, not discretisation, except where noted.

Claim	Arbiter	Residual
amp $\times$ phase ⁿ = bundled kernel	k_x -residue kernel	1.5×10^{-15}
residual <i>flat</i> in separation $n = 1 \dots 32$	—	—
left amplitude = x -reflection of right	parity identity	1×10^{-15}
lateral sweep = pairwise sum, distinct sources	$O(N^2)$ sum	1.5×10^{-13}
residual <i>flat</i> in lattice size	—	—
k_y -integrated kernel = closed form	Kupradze	falls to 2×10^{-5}
inter-plane sweep = pairwise sum	Kupradze	falls to 8×10^{-4}
lateral coupling does not wrap	kernel at true sep.	0
stratified operator reduces to whole space	k_z -residue kernel	1.1×10^{-15}
reverberation vanishes, uniform background	—	$< 10^{-15}$
self-energy $\equiv$ dressed $T_d = T_0(I - DT_0)^{-1}$	direct vs dressed	4×10^{-16}
reverberation obeys symplectic reciprocity	J_6 law, $\pm k$	8×10^{-14}
self-energy recedes with the reflector	—	$10^{-4} \rightarrow 10^{-12}$
partition: every pair once, self-term	support count	exact
GMRES = dense LU = Neumann, $\rho(G_0 T_0) = 0.92$	two direct methods	1.2×10^{-12}
ΔG_0 lies in the rank-3 one-way span	pseudo-inverse residual	4×10^{-9}
ΔG_0 magnitude, full 9×9 forward	Kennett reflectivity	2×10^{-10}
weak, strong and multi-interface, $k_{\parallel} h \leq 3$	—	—
P/S flux factor, predicted not fitted	$i\beta\sqrt{\eta_S}/\alpha\sqrt{\eta_P}$	8 s.f.
y -integral of 3-D form = 2½-D propagator	k_z residue vs Kupradze	4×10^{-13}
layered table reduces, uniform background	closed form	1.3×10^{-14}
residual <i>flat</i> in the transverse rule	—	—

Three points of method are worth recording because each cost time to learn. Symmetry and reciprocity invariants are homogeneous of degree one and therefore blind to an overall scale error; every such gate here is paired with one absolute comparison. Where a tolerance had to be loosened it was loosened only after measuring why, and the accompanying test asserts the

property that actually discriminates — that the residual is flat in separation, or flat in lattice size — since the dangerous failure is a factor that accumulates along the sweep. Finally, a residual that does *not* fall under refinement is a convention error, not a quadrature error: two of the discrepancies met here were exactly that, one an index ordering and one an attenuation mismatch between an operator and the Green’s function it multiplied.

On the self-energy, and which invariant applies to it. The same-depth reverberation ΔG_0 does *not* satisfy the whole-space symmetry $WM = (WM)^T$: measured 0.33 to 1.06 across k_x , at every separation. That is not a defect but a misapplied invariant. At coincident planes the *full* layered kernel fails the symplectic reciprocity law at 9.5×10^{-3} while ΔG_0 satisfies it at 8×10^{-14} : the non-reciprocal part is the one-sided jump carried by the direct term, and removing that term is precisely what (3) does. The reverberation is the clean reciprocal object; the whole-space symmetry is the wrong thing to demand of it. This is recorded because the same pattern — a plausible invariant that the exact reference also fails — has already cost this programme one false gate.

The absolute magnitude of ΔG_0 , now closed. Reciprocity, the vanishing without layering, the recession with the reflector and the dressed- T identity are all homogeneous of degree one and therefore say nothing about scale. The magnitude is settled instead by a *forward* comparison carrying no fitted quantity. At a plane over a layered half-space the reverberation must factor through the one-way mode maps of §2.4,

$$\Delta G_0 = D_{\uparrow} R_D S_{\downarrow}, \quad (5)$$

with R_D the total downward reflection matrix referenced at the plane — built by the Kennett reflectivity recursion alone, every internal multiple included. Predicting the full 9×9 from (5) and differencing it against the measured ΔG_0 leaves a relative residual of 2×10^{-10} , at weak contrast, at strong contrast, and on a fast/slow/fast stack, out to $k_{\parallel}h = 3$ in the evanescent regime.

Two things had to be measured rather than assumed for that comparison to mean anything. First, the part of ΔG_0 lying *outside* the rank-three one-way span of (5) is 4×10^{-9} or smaller, so the pseudo-inverse extractions used elsewhere in this programme are reading the operator and not a projection of it. Second, the mode-*conversion* channels $P \leftrightarrow SV$ are invisible to any per-mode ratio test, since a similarity $R \mapsto C^{-1}RC$ by a diagonal C leaves every diagonal entry exactly fixed. Compared directly they disagree — and the disagreement is precisely the Kennett P/S flux normalisation, $c = i\beta\sqrt{\eta_S} / \alpha\sqrt{\eta_P}$, which the mode bridge does not carry and the reflectivity recursion does. Written down *a priori* it reproduces the measured factor to eight significant figures, identically across all three models, and collapses the forward residual from 0.76 to the in-span floor. That is a convention reconciled, not a constant tuned.

The difference is not cosmetic. The mode bridge normalises polarisations to unit *displacement*, and bilinearly, so that they continue analytically into the evanescent regime. The reflectivity recursion normalises to unit *energy flux*, because that is what makes the interface matrices symmetric and unitary — $T_u = T_d^T$ — and without those identities the recursion is not valid. Vertical flux goes as $\rho v^2 \eta |A|^2$, so unit-flux amplitude goes as $1/(v\sqrt{\rho\eta})$, which is where the velocity and the $\sqrt{\eta}$ come from; the factor i on SV is the eigenvector convention of the recursion itself. This same similarity already appears in this package, converting the slab solver’s reflection matrix to the Kennett convention — the sweep solver needed a reconciliation the slab solver had already made, and the two agreeing is corroboration rather than coincidence.

6 The three-dimensional case, and why it is not six sweeps

The architecture above is a 2½-D result. Extending it to three dimensions by giving each sweep a transverse-wavenumber quadrature — the obvious reading, and the one this programme’s design

note originally took — does not work, and the reason is worth stating because it is quantitative rather than aesthetic.

A sweep carries a running accumulator indexed by its transverse quadrature nodes. In 2½-D that accumulator is $(n_z, n_{k_z}, 9)$, a few megabytes, because y is invariant and k_y is a single parameter. In three dimensions y becomes a lattice axis *and* the transverse rule becomes two-dimensional, so the accumulator is $(n_z, n_x, n_k, 9)$. At the converged transverse rule — $k_r p = 30$, $\Delta k = 0.156$, reaching 3.2×10^{-7} against the closed form — that is 68 GB on a 16^3 lattice with two live at once, and the inter-plane stack stored spectrally is 615 GB. Both were measured, not estimated.

The same operators *tabulated in real space* are 1.2 MB and 149 MB. So in three dimensions the coupling is split by depth separation rather than by direction:

$$G_0 = \underbrace{P(\Delta x, \Delta y, 0)}_{\text{closed form, exact}} + \underbrace{P(\Delta x, \Delta y, \Delta z)}_{\text{closed form}} + \underbrace{\Delta G_0(\Delta x, \Delta y, \Delta z)}_{\text{spectral, transformed once}}, \quad (6)$$

with the whole-space part taken from the closed-form propagator at every separation and *only* the layer reverberation transformed from (k_x, k_y) . That subtraction is the economy of the scheme and not a convenience: the full layered kernel inherits the whole-space kernel’s slow spectral decay and needs some 2.4×10^6 transverse nodes, whereas the reverberation converges at $k_r p = 10$ on about 6.6×10^4 — a factor of 36. The resulting build costs 2.1 h at 16^3 , once, outside the Krylov loop.

The (z, x) closed-form y -integral. The two architectures meet exactly, through the y -integral, and it is worth being precise because it is often stated as though 2½-D required a propagator of its own. It does not. For a y -invariant model every voxel is a line of identical voxels along y , so the coupling between two (z, x) columns is the three-dimensional propagator summed along that line,

$$P_{2\frac{1}{2}}(\Delta x, \Delta z; k_y) = \int_{-\infty}^{\infty} P(\Delta x, y, \Delta z) e^{-ik_y y} dy, \quad (7)$$

so the 2½-D propagator is the k_y component of the y -transform of the three-dimensional one, and at $k_y = 0$ it is literally its y -integral. There is no separate 2½-D closed form — only a slice of the three-dimensional one. Equation (7) is what licenses using the same closed-form table for both geometries, and it is the reason the $\Delta z = 0$ divergence discussed in §2.4 is a property of the k_z -pole construction rather than of the propagator itself.

This is measured, not asserted. Evaluating the left side by the k_z -residue kernel at $k_y = 0$ and inverse-transforming over k_x , against the right side by direct y -quadrature of the Kupradze closed form — two constructions with no derivation in common — agrees to 4×10^{-13} at $\Delta x = 0$ and 1.5×10^{-11} at two pitches (`gate_y_integral_2p5d`). The damping matters: at real ω both integrands decay only algebraically and neither route is clean.

Rebuilding three dimensions: the k_y quadrature. Equation (7) runs one way, from three dimensions to a slice. The step in the other direction is the one the solver actually performs, and it is the inverse transform,

$$P(\Delta x, y, \Delta z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} P_{2\frac{1}{2}}(\Delta x, \Delta z; k_y) e^{ik_y y} dk_y, \quad (8)$$

so a three-dimensional response is assembled by running the 2½-D construction once per k_y node and summing. Every sweep of §2.1 and every Kennett recursion is unchanged inside the integrand; k_y enters only as the parameter it already was. In practice (8) is a trapezoid on a uniform grid symmetric about $k_y = 0$, endpoints half-weighted, with the $1/2\pi$ carried on the quadrature weight.

Unlike the sweeps themselves, this step is quadrature-limited rather than exact, which is why the two rows it governs in §5 are reported as convergence and not as tolerances. Refining the rule

from $(k_{y\max}, n_{k_y}) = (60, 1201)$ to $(120, 3201)$ takes the k_y -integrated same-depth kernel from 2.3×10^{-4} to 2.3×10^{-5} against the Kupradze closed form (`gate_sweep_rung2_lateral`), and refining from $(60, 801)$ to $(120, 2401)$ takes the k_y -integrated inter-plane sweep from 5.2×10^{-3} to 8.0×10^{-4} against the real-space pairwise sum (`gate_sweep_rung3_vertical`). Both refinements raise the truncation $k_{y\max}$ and the node density together, so what is demonstrated is that the quadrature converges, not which of the two limits it; the cause of the slow convergence is the same algebraic decay of the whole-space kernel costed in the accumulator argument above, and it is the reason (6) sends only the reverberation through a transverse rule.

On the arbiter that was dropped. The validation ladder originally carried one further rung: the composed operator against `FFTProp`, the faithful port of the thesis’s own Fortran, as a convergence study rather than a tolerance. That rung is *dropped*, and the reasoning is worth stating because it is about what an arbiter is for.

`FFTProp` carries no T_0 and performs no multiple-scattering solve — it propagates given scattering sources — so the “cylinder Mie T_0 against cube T_0 ” shape difference the rung was built around does not arise. Its free surface does not reproduce the free-surface reflection: two independent constructions, the Aki & Richards displacement formula and the traction-free condition applied to the stratified eigenvectors, agree with each other to six decimal places and disagree with it, its $|R_{PP}|$ staying within 1% of unity across the whole propagating range where the true surface dips to 0.15, and its P – SV conversion reaching 10^{-3} where the true one reaches 0.63. And its state vector is carried in cylindrical harmonics — which are not a coordinate choice but the shape of its *cylinders* entering the representation, where this construction discretises into cubic voxels and carries the Cartesian nine-component state throughout.

The rung’s case was provenance. Set against four arbiters that already pass — Kennett reflectivity at 10^{-15} on the marine stack, the closed-form Kupradze propagator at 3×10^{-16} , the entirely different FFT-convolution architecture at 5×10^{-16} , and a dense LU with an explicit Neumann series at 10^{-12} — a fifth reached through a lossy change of representation, against a code with a different discretisation and a free surface that does not convert, would principally measure the difference between two discretisations. That is a real quantity, and it is not validation of the sweeps.

7 What is distinctive

Two choices set this apart from a standard FFT-accelerated volume integral equation. First, the forward operator is a directional partial-wave summation rather than a circular-convolution FFT, so the in-plane coupling is summed along the separated axes and the $\Delta z = 0$ divergence is structurally absent rather than regularised — and where the stratified background does contribute at equal depth, §2.4, it enters as a difference in which the divergence cancels rather than as a term requiring regularisation. *In three dimensions the first of these changes character:* by §6 the same-depth coupling is a direct real-space sum of the closed form rather than a directional summation, and what survives of the partial-wave structure is the depth direction and the reverberation. The absence of any periodicity assumption survives intact, and by a stronger route — a real-space table assumes none at all, where a transverse quadrature merely avoids introducing one. Second, the depth structure is carried exactly by the stratified propagator on the true layered background, not approximated on a homogeneous whole-space kernel. The multiple scattering is left where it belongs, in a Krylov solve that calls the sweep as its matvec, so the propagator stays a clean, stable, forward summation and the solver carries the contrast.

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